

Commutative Algebra

Day 2 → Atiyah Macdonald

• Extension & Contractions → $f: A \xrightarrow{\text{hom}} B$

then, if $\mathfrak{a} \subseteq A$, $\mathfrak{b} \subseteq B$, define $\mathfrak{a}^e \triangleq \mathfrak{b}^c$ as ideal generated by images of the generators of $\mathfrak{a}$ in B & $\mathfrak{b}^c$ as $f^{-1}(\mathfrak{b})$ respectively.

Propⁿ: if $\mathfrak{b}$ is prime in B , $f^{-1}(\mathfrak{b})$ is prime in A

Proof: Let $xy \in f^{-1}(\mathfrak{b})$. Then, $f(xy) \in \mathfrak{b}$.

As f is hom, hence $f(x) \cdot f(y) \in \mathfrak{b} \Rightarrow$ either $f(x) \in \mathfrak{b}$ or $f(y) \in \mathfrak{b} \Leftrightarrow x \in f^{-1}(\mathfrak{b})$ or $y \in f^{-1}(\mathfrak{b})$.

Hence $f^{-1}(\mathfrak{b})$ is prime.

(Also if f were surjective, $f(\mathfrak{a})$ is an ideal in B)

But then, do prime $\mapsto$ prime? Correspondence Thm
if $\mathfrak{a} \supseteq \ker f$, then works.

Otherwise, counterexample in $f: \mathbb{Z} \rightarrow \mathbb{Z}_2 \rightarrow \text{set } \mathfrak{a} = (3)$
 $n \mapsto n \pmod{2}$
 $\therefore \ker f = 2\mathbb{Z} \not\subseteq (3)$

Exercises

(11) $x^2 = x \quad \forall x \in A$.

(a) $2x = 0 \quad \forall x \in A$

Pf: $(a+a)^2 = a+a = 2a \mid \rightarrow 2a^2 + 2a = 2a \Rightarrow 2a^2 = 0 \Rightarrow 2a = 0 \quad \forall a$

(b) Let $\mathfrak{p}$ be a prime ideal in A .

$A/\mathfrak{p}$ must be a domain $\rightarrow R = \{x + \mathfrak{p} : x \in A\}$
is a domain, let image of x be $\bar{x}$.

Claim: R is a field.

Proof: Let $\bar{x} \neq 0$ in R .

As $\bar{x}^2 = \bar{x}$ in R , & R is a domain, hence

$$\bar{x}(\bar{x} - 1) = 0 \Rightarrow \bar{x} = 1 \text{ or } \bar{x} = 0$$

Thus, as either $\bar{x} = 0$ or $\bar{x} = 1 \forall x \in A$, hence
 if $\bar{x} \neq 0$, then $\bar{x}$ is a unit, thus R is a field.
 As R has only $\bar{0}$ & $\bar{1}$, hence $R \cong \mathbb{F}_2$, thus
 R has only two elements.

(iii) Let $I = (a_1, a_2, \dots, a_n)$ be an ideal in A .

Induct $\rightarrow$ suffices to show for (x, y) is principal.
 $\rightarrow x + y + vx$ is the generator $\rightarrow$ check

(12) Local Ring A contains no idempotent $\neq 0, 1$.

Pf:- Only one maximal ideal $\mathfrak{m}$.

Suppose $\exists x \in A \ni x^2 = x, x \neq 0, 1$.

Then, note that x is not a unit (as else $x^2 x^{-1} = x x^{-1} \Rightarrow x = 1$)

Thus, x must be in $\mathfrak{m}$. (As if $x \in A - \mathfrak{m}$, $(x) = (1)$)

(Hence all idempotent elements in $\mathfrak{m}$).

But as $x(x-1) = 0$, & $J(A) = \mathfrak{m}$ as A is local,

hence as $x \in \mathfrak{m}$, $1-x$ is a unit, however

$1-x$ is a zero divisor, contradiction!

(14) $\Sigma \rightarrow$ ^{collection} all ideals ^{in A} where every element is a zero divisor. Let $<$ be the partial order by inclusion.

& let $I_1 \subsetneq I_2 \subsetneq I_3 \subsetneq \dots$ be a chain.

Then, $\bigcup I_n$ is an ideal where every element is a zero divisor & hence an upper bound, thus by Zorn's lemma, Σ has a maximal element.

Let $\mathfrak{m}$ be 'a' maximal element.

Claim:- $\mathfrak{m}$ is prime.

Proof:- Suppose not, i.e., $\exists x, y \notin \mathfrak{m}$ but $xy \in \mathfrak{m}$.

Consider (x, m) & (y, m) . As m is a maximal,
hence (x, m) & $(y, m) \notin \Sigma$, thus, in (x, m) , $\exists$
a non-zero divisor a & in (y, m) , b .

write $a = x t_1 + m_1 u_1$, $b = y t_2 + m_2 u_2$ for $m_1, m_2 \in m$
& $t_1, t_2, u_1, u_2 \in A$

however, $ab = x y t_1 t_2 + x m_2 u_2 + y m_1 u_1 t_2 + m_1 m_2 u_1 u_2 \in m$.
as $xy \in m$.

however, as a & b are not zero divisors, so is ab ,
but as $ab \in m$, it's a zero divisor, a contradiction.
Hence m is prime.

Maximal Element of Zorn's Lemma & Primality
This approach.

(15) (ii) Trivial, if $1 \in p$ then $p = A$, & not prime.

(i) if $\bar{a} = \langle E \rangle$, then $v(E) = v(\bar{a}) = v(\sqrt{\bar{a}})$ (Fundamental AG prop)

Pf: let $p \in v(E)$ hence, $p \supseteq E$, & as $\bar{a} = \langle E \rangle$, hence
 $p \in v(\bar{a})$ as well, thus $v(E) \subseteq v(\bar{a})$.

Conversely, let $p \supseteq \bar{a}$, i.e., $p \supseteq \langle E \rangle$.

Claim: $p \supseteq E$

Proof: Suppose that $p \not\supseteq E$. Pick $x \in E$ but $x \notin p$.

As $p \supseteq \langle E \rangle$, & $x \in \langle E \rangle$ but $x \notin p$, this
is a contradiction, hence $p \supseteq E$.

Thus, $p \in v(E)$. Hence, $v(\bar{a}) \subseteq v(E)$.

Hence, $v(E) = v(\bar{a})$.

Now, we'll show that $v(\bar{a}) = v(\sqrt{\bar{a}})$

Let $p \supseteq \sqrt{\bar{a}}$. As $\bar{a} \subseteq \sqrt{\bar{a}}$, hence $p \supseteq \bar{a}$,
hence $v(\sqrt{\bar{a}}) \subseteq v(\bar{a})$.

Let $p \supseteq \bar{a}$. Let $x \in \sqrt{\bar{a}}$, hence $\exists n > 0$ s.t.
 $x^n \in \bar{a} \in p$. As p is prime, hence $x \in p$.

Thus, $\sqrt{a} \subseteq p$, thus, $v(a) \subseteq v(\sqrt{a})$.

Hence, we have $v(a) = v(\sqrt{a}) = v(E)$.

15)(iii) $\{E_i\}_i$ any family of sets in ring A , then

$$v\left(\bigcup_i E_i\right) = \bigcap_i v(E_i)$$

Pf $\div$ let prime $p \supseteq \bigcup_i E_i$. Thus, $\forall i, E_i \subseteq p$.

Hence, $p \in v(E_i) \forall i$, hence $p \in \bigcap_i v(E_i)$.

Hence, $v\left(\bigcup_i E_i\right) \subseteq \bigcap_i v(E_i)$

Conversely, let $p \in \bigcap_i v(E_i) \Rightarrow p \in v(E_i) \forall i$

hence, $\forall i, p \supseteq E_i$ hence $p \supseteq \bigcup_i E_i$
 $\Rightarrow p \in v\left(\bigcup_i E_i\right)$

thus, $\bigcap_i v(E_i) \subseteq v\left(\bigcup_i E_i\right)$

thus, $\bigcap_i v(E_i) = v\left(\bigcup_i E_i\right)$

(iv) For $a, b \in A$, we have —

$$v(ab) = v(a \cap b) = v(a) \cup v(b)$$

Proof $\div$ Let $p \in v(ab)$, i.e., $p \supseteq ab$.

Then, $p \supseteq a$ or $p \supseteq b$, hence $p \in v(a)$ or $p \in v(b)$

hence, $p \in v(a) \cup v(b)$, thus,

$$v(ab) \subseteq v(a) \cup v(b)$$

Conversely, let $p \in v(a) \cup v(b)$, i.e., $p \supseteq a$ or $p \supseteq b$

Thus, as p is prime, hence WLOG $p \supseteq a$.

Let $x \in b$. Then, $ax \subseteq p$, hence as x was arbitrary $ab \subseteq p$, thus, $p \in V(ab)$, hence $V(a) \cup V(b) \subseteq V(ab)$.
Thus, $V(ab) = V(a) \cup V(b)$.

As $ab \subseteq a \cap b$, hence if $p \in V(a \cap b)$,
i.e. $a \cap b \subseteq p$, thus $ab \subseteq p \Rightarrow p \in V(ab)$
 $\therefore V(a \cap b) \subseteq V(ab)$.

Now, let $p \in V(ab)$, i.e. $p \supseteq ab$. As p is prime,
 $p \supseteq a$ or $p \supseteq b$, thus, $a \cap b \subseteq p$, hence $p \in V(a \cap b)$
hence $V(ab) \subseteq V(a \cap b)$.

Thus, we have $V(ab) = V(a \cap b) = V(a) \cup V(b)$.

1b) (a) $\text{Spec}(\mathbb{Z}) \rightarrow \mathbb{Z}$ is a PID, hence any prime ideal
in $\mathbb{Z}$ is (p) where p is prime.

So, given $E \subseteq \mathbb{Z}$, $V(E) \supseteq (p) \Rightarrow p$ divides
every element in E . Hence any closed set in $\text{Spec}(\mathbb{Z})$
is Vanishing point of elements all sharing a common
prime factor.

Final $\downarrow$ $V(0) = \text{Spec } \mathbb{Z}$ & $V(n) \forall n \neq 0$ is closed,
as if $n = \prod_{1 \leq i \leq k} p_i^{\alpha_i}$ then its just in the ideal

generated by $E = \{p_1, p_2, \dots, p_k\}$.

(b) $\text{Spec}(\mathbb{R}) \rightarrow$ Given $E \subseteq \mathbb{R}$, if $p \in V(E)$,
as $\mathbb{R}$ is a field, $p = (0)$. Thus,
 $(0) \supseteq E$, $E = \{0\}$ or $E = \emptyset$.

Bonus: Let k be a field, $\text{Spec} \left(\frac{k[x]}{(x^2)} \right) = ?$

Let $R = \frac{k[x]}{(x^2)}$. As k is a field, $k[x]$

If $\mathfrak{p}$ is a prime ideal in R , then by Correspondance theorem, $\mathfrak{p}$ corresponds to a prime ideal in $k[x]$ containing (x^2) , say $\mathfrak{m}$.

Then, as $\mathfrak{m} \ni x^2$, & $\mathfrak{m}$ is prime, thus $x \in \mathfrak{m}$ & hence $(x) \subseteq \mathfrak{m}$, however as (x) is maximal, it must be that $(x) = \mathfrak{m}$.

$\therefore$ The corresponding prime ideal in $\frac{k[x]}{(x^2)}$

is $(\bar{x}) = \mathfrak{m}'$, hence $\text{Spec} \left(\frac{k[x]}{(x^2)} \right) = \{ \mathfrak{m}' \}$

(Imp note $\rightarrow$ if analysing $\text{Spec}(k \times k) \rightarrow$ then $(0,0) = (1,0) \times (0,1)$
hence as $(0,0) \in \mathfrak{p}$, hence either $(1,0) / (0,1) \in \mathfrak{p}$,
but then $(k,0) / (0,k) \in \mathfrak{p}$, & hence
 $\mathfrak{p} = I \times k$ or $k \times I$ for prime I , i.e.
 $(0) \times k$ or $k \times (0)$.)

Bonus 2

$\text{Spec} \left(\frac{\mathbb{R}[x]}{(x^2-x)} \right) ?$

$\hookrightarrow$ Let $\mathfrak{p}$ be prime ideal in $\frac{\mathbb{R}[x]}{(x^2-x)}$, then, by correspondance

theorem, $\mathfrak{p} \leftrightarrow \mathfrak{p}'$ in $\mathbb{R}[x] \ni \mathfrak{p}' \supseteq (x^2-x)$.

As $x(x-1) \in \mathfrak{p}'$, hence either $x \in \mathfrak{p}'$ or $x-1 \in \mathfrak{p}'$.

Case 1: $x \in \mathfrak{p}' \Rightarrow (x) \subseteq \mathfrak{p}'$. as (x) is maximal ideal in $\mathbb{R}[x]$,
hence $\mathfrak{p}' = (x)$, & thus, $(\bar{x})$ is the only element in

$\text{Spec} \left(\frac{\mathbb{R}[x]}{(x^2-x)} \right)$.

Case 2: $x-1 \in p' \Rightarrow (x-1) \subseteq p'$ & again, as $(x-1)$ is maximal, hence $(\overline{x-1})$ is the only element in $\text{Spec} \left(\frac{\mathbb{R}[x]}{(x^2-x)} \right)$.

closed sets in
Thus, $\uparrow \text{Spec} \left(\frac{\mathbb{R}[x]}{(x^2-x)} \right) = \{ \overline{(x-1)}, \overline{(x)} \}, \{ \overline{(x-1)}, \overline{(x)} \}$.

Key: $p_1 \cap p_2$ corresponds to a smaller ideal $\Rightarrow$ lesser generators $\Rightarrow$ lesser constraints, & hence $V(p_1 \cap p_2)$ is potentially larger (more freedom). (Treat prime ideals as points in Zariski Topology $\rightarrow$ to analyse them)

Propⁿ: $\text{Spec}(A \times B) \cong \text{Spec}(A) \amalg \text{Spec}(B)$

Proof: Let p be any prime ideal in $A \times B$.

Then, as $(0,0) = (1,0) \cdot (0,1)$, hence $(0,1) \in p$ or $(1,0) \in p$.

If $(0,1) \in p$, this means $(0) \times B \subseteq p$.

By Correspondance Theorem, this means p is in correspondence with prime ideals in $\frac{A \times B}{(0) \times B} \cong A$.

So, prime ideals of $A \times B$ containing $(0,1) \leftrightarrow q \times B$, q prime in A .

Similarly, prime ideals of $A \times B \ni (1,0) \leftrightarrow A \times q$, q prime in B .

As $q \times B \cap A \times q' = \{(0,0)\}$, & if $I \subseteq p$ in $A \times B$, then $I = I_A \times I_B$ (always possible), hence for $I_A \times I_B \supseteq q \times B$, say, as $I_B \supseteq B$ trivially this just means $I_A \supseteq q$, & hence we can say $\text{Spec}(A \times B) \cong \text{Spec}(A) \amalg \text{Spec}(B)$.

(Back) $\rightarrow$ (.) $\text{Spec}(\mathbb{C}[x]) \rightarrow \mathbb{C}$, as $\mathbb{C}$ is algebraically closed, hence if p is prime in $\mathbb{C}[x]$, it is maximal & by Hilbert, $\exists a \in \mathbb{C} \ni p = (x-a)$.

so for $E \subseteq (x-a)$, we need all elements of E to have 'a' as zeroes, i.e. if $f \in E$, $f(a) = 0$.

Δ $\text{Spec}(\mathbb{C}[x]) = \{(x-a) : a \in \mathbb{C}\} \cup \{(0)\}$.

If $f \equiv 0$, then $V((0)) = \text{Spec}(\mathbb{C}[x])$

If $f \equiv c$ for $c \in \mathbb{C}$, then as $\mathbb{C}$ is a field, hence we want $V((1))$, however, this is $\emptyset$.

If f is a non-constant poly, then $f(x) = \prod_{1 \leq i \leq k} (x-a_i)^{r_i}$

If $f \in p$ for prime (hence maximal p) thus $p \neq (0)$

$(x-a_i) \in p$ for any i . Hence, we want

$$\bigcup_{1 \leq i \leq k} V(x-a_i) = V\left(\bigcap_{1 \leq i \leq k} (x-a_i)\right)$$

(.) $\text{Spec}(\mathbb{R}[x]) \rightarrow$ Let p be prime in $\mathbb{R}[x]$. Again, it is maximal. Thus $p = (f)$ or $p = (g)$ where

$\deg f = 1$ & $\deg g = 2$. Hence,

$$\text{Spec}(\mathbb{R}[x]) = \underbrace{\{(f), (g) : \deg f = 1, \deg g = 2, g \text{ irreducible}\}}_{\cup \{(0)\}}$$

if $g(x) = ax^2 + bx + c$, then $b^2 - 4ac < 0$

(.) $\text{Spec}(\mathbb{Z}[x]) \rightarrow$ Let p be prime in $\mathbb{Z}[x]$.

that that any $(p) \ni p$ is prime is a prime ideal. Consider $\phi_p: \mathbb{Z}[x] \rightarrow \mathbb{Z}_p[x]$,
 $f(x) \mapsto f(x) \pmod{p}$

As $\mathbb{Z}_p$ is a field, hence $\mathbb{Z}_p[x]$ is a P.I.D, thus,

any prime ideal in $\mathbb{Z}_p[x]$ is maximal, & as

any maximal ideal in a P.I.D is generated by irreducibles,

hence m prime in $\mathbb{Z}_p[x] \Rightarrow m = (f(x))$ for some f irreducible in $\mathbb{Z}_p$.

(Now, we know that f irred in $\mathbb{Z}_p \Rightarrow f$ irred in $\mathbb{Z}$)
 thus, f irred in $\mathbb{Z}$.)

So, by correspondence theorem, the prime ideals in $\mathbb{Z}[x]$ correspond to prime ideals in $\mathbb{Z}_p[x]$ containing (p) . Hence, we need as f is irred in $\mathbb{Z}[x]$, the second type of prime ideals in $\mathbb{Z}[x]$ are (p, f) where p is prime & f is irreducible in $\mathbb{Z}[x]$.

Also, if f is irred in $\mathbb{Z}[x]$, then (f) is a prime ideal as $\frac{\mathbb{Z}[x]}{(f)}$ is an integral domain.

thus, $\text{Spec}(\mathbb{Z}[x]) = \{ (p), (p, f), (f) : p \text{ is prime \& } f \text{ is irred in } \mathbb{Z}[x] \} \cup \{0\}$

(17) For $f \in A$, let $X_f = V(f)^c$. Then, $B = \{X_f : f \in A\}$ is a basis for Zariski Topology. hence $p \in X_1$

Proof:- (i) Let $p \in \text{Spec } A$, as $\text{Spec } A = V(1)^c$ (as $1 \notin p \forall p$ prime in A)

(ii) Suppose $X_f \cap X_g \neq \emptyset$. Let $p \in X_f \cap X_g = V(f)^c \cap V(g)^c = (V(f) \cup V(g))^c$

$$\text{As } V(f) \cup V(g) = V((f)) \cup V((g)) = V((f) \cap (g)) = V((fg))$$

$$\therefore p \in V((f) \cap (g))^c = V((fg))^c = X_{fg}$$

$$\text{thus, } X_f \cap X_g \subseteq X_{fg}$$

$$\text{Let } p \in X_{fg}, \text{ i.e., } p \in V((fg))^c \Rightarrow p \in V((f) \cap (g))^c$$

$$\Rightarrow p \in (V((f)) \cup V((g)))^c = V((f))^c \cap V((g))^c$$

$$= X_f \cap X_g$$

$$\therefore X_{fg} \subseteq X_f \cap X_g$$

Thus, we have $X_f \cap X_g = X_{fg}$, & hence as X_{fg} is an element of B , hence, we have shown the gluing property, & hence B is a basis for the Zariski Topology.

Claim:- The topology generated by B is precisely the Zariski Topology.

Proof:- T.S:- For any open $U \in \text{Spec } A$, as U^c is closed, hence $U^c = V(E)$ for some $E \subset A$.

Then $E = \bigcup_{a_i \in E} \{a_i\}$. Then, $V(E) = V\left(\bigcup_{a_i \in E} \{a_i\}\right)$

$$= \bigcap_{a_i \in E} V(\{a_i\})$$

$$\therefore U = \left(\bigcap_{a_i \in E} V(\{a_i\}) \right)^c = \bigcup_{a_i \in E} V(\{a_i\})^c$$

$$= \bigcup_{a_i \in E} X_{a_i}$$

so, U is a union of basic elements, hence topology generated by B is precisely the Zariski Topology.

(ii) $X_f = \emptyset \Rightarrow f$ is nilpotent.

Proof:- This means $V(f) = \text{Spec } A$, i.e.,

$f \in \mathfrak{p} \ \forall$ prime $\mathfrak{p}$, & hence $f \in \bigcap_{\mathfrak{p} \text{ prime}} \mathfrak{p} = \text{Nil}(A)$,

& hence f is nilpotent.

(iii) $X_f = X \Rightarrow f$ is a unit. $\nearrow \text{Spec}(A)$

Proof:- This means $V(f)^c = X \Rightarrow V(f) = \emptyset$.

Hence, there is no prime ideal containing f .

However, if f were not a unit, by Krull,

$f \in \mathfrak{m}$ for some maximal ideal $\mathfrak{m}$ in A , but $\mathfrak{m}$ is prime $\Leftrightarrow \Leftrightarrow$ hence, f must be a unit.

(iv) $X_f = X_g$ iff $\sqrt{(f)} = \sqrt{(g)}$.

Proof:

Claim: for any ideal I , $\sqrt{I} = \bigcap_{P \supseteq I} P$

Proof:

note that $\sqrt{I} = \phi^{-1}(\text{Nil}(A/I))$

where $\phi: A \rightarrow A/I$ is the canonical homomorphism.

Now, we know $\text{Nil}(A/I) = \bigcap_{\substack{P' \text{ prime} \\ \text{in } A/I}} P'$

By Correspondance Theorem, $P' \text{ in } A/I \iff P \text{ in } A$
 $\exists P \supseteq I$.
 as $\ker \phi = I$.

hence $\sqrt{I} = \phi^{-1}\left(\bigcap_{\substack{P' \text{ prime} \\ \text{in } A/I}} P'\right) = \bigcap_{\substack{P \text{ prime} \\ P \supseteq I}} P$.

Using the claim, we have $\sqrt{(f)} = \bigcap_{P \supseteq (f)} P$.

$(\Rightarrow)$ Suppose, $V((f)) = V((g))$, hence $\forall P \supseteq (f)$,
 we know $P \supseteq (g)$ as well, hence

$\sqrt{(f)} = \bigcap_{P \supseteq (f)} P = \bigcap_{P \supseteq (g)} P = \sqrt{(g)}$. Thus, $\sqrt{(f)} = \sqrt{(g)}$.

$(\Leftarrow)$ Suppose $\sqrt{(f)} = \sqrt{(g)}$.

Let $P \in V((f))$. Then, this means, $(f) \subseteq P$.

Let $x \in (g)$. As $\sqrt{(g)} = \sqrt{(f)}$, & $x \in \sqrt{(g)}$,
 hence $x \in \sqrt{(f)} \Rightarrow \exists n > 0 \ni x^n \in (f) \subseteq p$
 $\Rightarrow x \in p \Rightarrow (g) \subseteq p$, hence,
 $p \in V((g))$ as well, i.e., $V((f)) \subseteq V((g))$

As the argument is totally symmetric, hence, $V((g)) \subseteq V((f))$,
 & thus, $V((f)) = V((g))$.

Hence, $X_f = X_g \Leftrightarrow \sqrt{(f)} = \sqrt{(g)}$.

↳ In ($\Leftarrow$) could've directly used that $\sqrt{p} = p$,
 as p is prime, much cleaner.

Summary

(i) Primarily Zariski Topology ($\text{Spec } A$), its basis ($V(f), f \in A$),
 equality b/w its algebra & geometry $\rightarrow$ primarily $\rightarrow$
 • $V(E) = V(\langle E \rangle)$ & $V(I \cap J) = V(I) \cup V(J)$,
 • $V(\bigcup_{i \in I} E_i) = \bigcap_{i \in I} V(E_i) \rightarrow$ asserting that $V(E)$'s are
 arbitrary index set I precisely the closed sets.

(ii) • Review of $\rightarrow$ • Zorn's maximal, primality $\rightarrow$ create new
 ideals & multiply for contradiction!

• $\text{Spec}(A)$ for various A 's,

• $\sqrt{I} = \bigcap_{\substack{p \text{ prime} \\ p \supseteq I}} p$ & why its exactly same logic as
 $\text{Nil}(A) = \bigcap_{p \text{ prime}} p$

• Extensions & Contractions.